

GRIVITA, Romania

WMO: 154050

Lat: 44.7408N

Lon: 27.2947E

Elev: 169

StdP: 14.61

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	14.0	6.0	7.5	11.1	10.4	9.4	16.1	39.0	22.4	32.3	24.3	7.5	230	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	92.3	71.8	89.8	70.4	87.4	69.6	74.4	87.5	72.8	85.4	71.3	83.3	6.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.5	113.2	79.1	68.9	107.0	77.4	67.4	101.5	76.0	38.0	87.1	36.5	86.1	35.2	83.1	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.2	22.4	18.5	DB	2.7	96.6	4.2	3.2	-0.4	98.9	-2.9	100.8	-5.2	102.7	-8.3	105.0
			WB	2.2	78.0	4.2	3.1	-0.8	80.2	-3.3	82.0	-5.6	83.7	-8.7	86.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.1	29.2	35.6	42.8	52.8	63.0	71.0	73.7	73.5	65.3	53.3	42.7	32.7	(6)
(7)		DBStd	17.21	8.11	8.94	8.34	7.40	6.06	5.00	4.45	5.39	7.55	7.46	8.48	8.33	(7)
(8)		HDD50	2192	644	406	246	56	2	0	0	0	0	51	249	538	(8)
(9)		HDD65	5243	1109	823	687	367	109	10	1	5	93	367	669	1003	(9)
(10)		CDD50	3308	0	4	23	141	404	631	733	728	460	154	30	0	(10)
(11)		CDD65	882	0	0	0	2	45	191	269	268	102	5	0	0	(11)
(12)		CDH74	9069	0	0	4	77	578	1873	2688	2786	991	71	1	0	(12)
(13)		CDH80	3441	0	0	0	12	138	653	1079	1220	326	13	0	0	(13)
(14)	Wind	WSAvg	7.2	8.1	8.1	8.9	8.1	6.9	6.1	5.8	6.5	6.4	6.6	6.7	8.1	(14)
(15)	Precipitation	PrecAvg	19.5	1.2	0.8	1.2	1.3	2.3	2.7	1.8	1.6	2.0	1.6	1.4	1.6	(15)
(16)		PrecMax	28.0	3.3	2.2	2.8	3.6	6.1	4.5	4.3	3.8	5.8	3.1	4.1	4.4	(16)
(17)		PrecMin	12.6	0.1	0.0	0.0	0.2	0.4	0.5	0.3	0.4	0.0	0.1	0.0	0.1	(17)
(18)		PrecStd	4.0	0.8	0.6	0.7	0.7	1.4	1.0	1.0	1.0	1.4	0.9	0.9	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	65.1	72.4	81.4	89.0	92.6	95.1	95.4	90.8	81.7	70.8	54.7	(19)
(20)			MCWB	46.2	54.6	56.2	63.8	68.9	73.0	74.3	73.9	67.7	66.5	58.7	50.4	(20)
(21)		2%	DB	45.6	57.5	66.1	75.4	83.6	89.1	92.0	92.5	86.8	74.8	64.3	50.3	(21)
(22)			MCWB	43.0	50.4	53.0	60.3	66.5	70.9	72.2	71.7	66.7	62.3	56.7	46.9	(22)
(23)		5%	DB	42.2	52.5	61.3	71.2	80.0	86.7	89.2	89.9	83.7	70.3	58.6	46.2	(23)
(24)			MCWB	40.7	46.8	50.4	57.9	64.5	70.2	70.9	70.3	65.4	59.5	53.0	43.9	(24)
(25)		10%	DB	38.9	47.5	56.6	67.6	76.7	83.9	86.1	87.2	80.0	66.0	54.8	42.6	(25)
(26)			MCWB	37.6	44.1	47.7	56.0	62.2	69.1	69.8	69.4	64.4	57.8	51.4	40.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.5	55.2	57.6	65.3	70.4	75.3	76.1	77.0	71.1	68.0	59.7	52.1	(27)
(28)			MCDWB	49.2	63.8	70.4	79.0	84.3	88.5	89.8	90.5	83.2	78.8	67.7	54.8	(28)
(29)		2%	WB	43.5	50.8	53.7	61.8	68.0	73.3	74.3	74.3	69.3	63.5	56.6	47.2	(29)
(30)			MCDWB	45.3	57.1	64.0	72.7	81.3	85.2	87.1	88.1	80.4	71.9	62.8	49.9	(30)
(31)		5%	WB	40.7	47.3	51.2	59.2	66.0	71.7	72.7	72.5	67.6	61.0	54.1	44.1	(31)
(32)			MCDWB	42.3	51.4	60.1	69.5	76.9	84.0	85.0	85.6	78.2	68.5	57.9	46.6	(32)
(33)		10%	WB	37.8	44.5	48.5	57.0	63.8	70.1	71.2	70.7	65.9	58.7	51.6	40.9	(33)
(34)			MCDWB	38.9	47.5	55.3	65.7	73.9	81.5	82.5	83.0	76.9	64.8	54.4	42.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.4	13.8	17.9	21.2	22.1	21.5	21.5	22.1	21.9	18.6	13.2	11.0	(35)
(36)			MCDBR	14.1	23.5	27.8	28.6	27.0	25.8	26.8	26.3	27.0	25.6	21.3	15.9	(36)
(37)			MCWBR	10.9	15.7	16.3	15.0	12.2	10.2	10.3	10.6	10.9	13.4	12.9	12.2	(37)
(38)		5% WB	MCDWB	13.5	21.5	26.2	26.5	25.5	24.1	23.1	23.1	24.2	23.1	18.6	15.8	(38)
(39)	MCWBR		10.9	15.1	15.9	14.9	12.7	10.3	9.9	10.9	10.7	13.0	12.1	12.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.344	0.373	0.406	0.461	0.434	0.437	0.451	0.465	0.430	0.392	0.366	0.346	(40)	
(41)		taud	2.296	2.221	2.182	2.042	2.203	2.226	2.195	2.145	2.206	2.298	2.320	2.312	(41)	
(42)		Ebn at Noon	238	251	259	254	266	265	259	250	249	242	227	223	(42)	
(43)		Edh at Noon	27	34	41	51	44	44	45	45	39	31	26	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	421	682	1060	1480	1868	2009	2036	1824	1325	849	466	354	(44)	
(45)		RadStd	61	97	110	151	132	137	128	106	99	95	54	47	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (57 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	-221	+126	+79	

Nomenclature: See separate page